Dual-Branch Selective Multiscale Fusion Network for Hyperspectral and LiDAR Classification

Minghui Wu, Shuo Shi, Youchai Lin, Deping Chu, Jiayun Niu, Xingtao Tang, and Wei Gong

***Abstract*—The joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) data is a fundamental task in remote sensing, benefiting from the strong complementarity between rich spectral information and 3-D structural characteristics. However, existing deep-learning-based fusion methods commonly rely on fixed receptive fields and predefined multiscale designs, which limits their adaptability to large-scale variations of ground objects. To address these challenges, this study proposes a dual-branch selective multiscale fusion network (DSMF-Net) for HSI–LiDAR joint ground-object classification. DSMF-Net adopts a dual-branch architecture to separately model spectral–spatial features from HSI and structural information from LiDAR, while preserving spatial correspondence between modalities. A selective multiscale convolution (SMConv) module is designed to enable competitive scale selection across multiple receptive fields, allowing the network to adaptively capture discriminative structures of varying spatial scales. In addition, a modality reconstruction constraint is incorporated to regularize multimodal representation learning, and a channel stitching and channel attention (CSCA) module is introduced to effectively fuse multiscale and multimodal features. Extensive experiments on the multimodal remote sensing Houston benchmark dataset demonstrate the superior performance and robustness of the proposed method. The results confirm that effective exploitation of multiscale information significantly enhances HSI–LiDAR feature fusion for fine-grained ground-object classification.**

***Index Terms*—Feature fusion, ground-object classification, hyperspectral image (HSI), light detection and ranging (LiDAR), multiscale feature learning.**

I. Introduction

GROUND-OBJECT classification is a fundamental task in remote sensing analysis and plays a critical role in applications such as national land surveys, ecological monitoring, precision agriculture, and urban planning, providing essential information for understanding surface composition and spatial patterns [1]. With the rapid advancement of remote sensing sensors, the synergistic fusion of multisource data has become an effective means to improve the accuracy and robustness of fine-grained ground-object classification [2]. Among various data sources, hyperspectral image (HSI) and light detection and ranging (LiDAR) are particularly representative due to their strong complementarity [3], [4], [5], [6], [7], [8]. HSI offers continuous and high-resolution spectral information, enabling detailed characterization of material composition and subtle spectral variations of ground objects [9], [10]. In contrast, LiDAR provides accurate 3-D structural information, allowing the acquisition of geometric and structural attributes [11]. As a result, HSI and LiDAR exhibit strong information-level complementarity, and their fusion to fully exploit respective advantages for improved ground-object classification has attracted increasing research interest [12], [13], [14]. However, due to the inherent differences in physical properties, semantic representations, and spatial scale distributions between heterogeneous data, most existing deep-learning methods fail to simultaneously extract discriminative multiscale features and effectively exploit the information of HSI and LiDAR, thereby limiting the performance of joint ground-object classification [15]. Moreover, real-world scenes are characterized by substantial scale variability: large-scale objects such as rivers and farmland coexist with small-scale structures such as roads, rooftops, and tree crowns. This wide range of spatial scales makes it difficult for convolutional kernels with fixed receptive fields to simultaneously capture fine-grained details and large-scale structural patterns. In HSI–LiDAR fusion tasks, such scale mismatch, together with modal heterogeneity, is further exacerbated, posing a major challenge to unified representation learning and limiting classification performance.

A variety of methods have been developed for the joint classification of HSI and LiDAR data. Early studies based on conventional machine learning techniques, such as morphological feature fusion, low-rank representation, and graph-based models, have promoted the development of HSI–LiDAR classification to a certain extent [16], [17], [18], [19], [20], [21]. However, these approaches are largely empirical and rely heavily on handcrafted features and extensive parameter tuning, which limits their scalability and generalization capability [22]. With the rapid progress of deep learning, data-driven methods have attracted increasing attention due to their strong ability to automatically learn hierarchical feature

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Minghui Wu, Shuo Shi, and Wei Gong are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, Hubei 430079, China (e-mail: wuminghui@whu.edu.cn; shishuo@whu.edu.cn; weigong@whu.edu.cn).

Youchai Lin, Jiayun Niu, and Xingtao Tang are with the School of Electronic Information, Wuhan University, Wuhan, Hubei 430079, China (e-mail: lyc20010101@163.com; niujiayun@whu.edu.cn; whutxt@whu.edu.cn).

Deping Chu is with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, Hubei 430079, China (e-mail: chudeping_2019@cug.edu.cn).

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representations [23], [24], [25]. Among them, convolutional neural networks (CNNs), benefiting from local receptive fields and parameter-sharing mechanisms, have become one of the most effective tools for HSI–LiDAR fusion-based classification tasks [26]. Existing CNN-based approaches typically adopt dual-branch architectures, in which spectral–spatial features of HSI and structural features of LiDAR are extracted by separate subnetworks and then integrated through feature concatenation or encoder–decoder-based multilevel fusion strategies, achieving notable performance improvements in multisource remote sensing scenarios. For instance, TBCNN [2] employs a dual-stream architecture to separately learn spectral and spatial–structural features and concatenates the outputs of the two branches for final classification. EndNet [27] further introduces parallel encoders to extract multimodal features and progressively aligns the information from different modalities in the fusion module, leading to enhanced multimodal feature representation. However, these methods generally rely on fixed-size convolution kernels and lack explicit multiscale feature modeling, making it difficult to fully exploit the complementary information of HSI and LiDAR when significant scale variation and modal heterogeneity coexist.

To enhance the model's ability to handle scale variations, existing studies commonly rely on predefined fixed multiscale structures. Typical approaches include InceptionNets that extract features with different convolution kernel sizes in parallel [28], [29], [30], [31], [32], atrous spatial pyramid pooling (ASPP) modules that construct multiscale receptive fields [33], [34], [35], and feature pyramid networks (FPNs) that fuse multiscale semantic features through downsampling and upsampling [36]. For instance, Gao et al. [37] proposed AMSSE-Net, which extracts spectral–spatial features at multiple scales through a predesigned multibranch convolution structure and captures LiDAR structural information with shallow CNNs. However, despite improving scale adaptability, these methods rely on fixed-scale configurations, making it difficult to dynamically adjust the receptive field according to the content of the input. When ground-object types have significant scale differences or there are large modal heterogeneities, fixed-scale combinations fail to capture the optimal multiscale feature expression. Furthermore, multibranch convolutions can only linearly combine or weight features, lacking competition and selection mechanisms, which limits the ability to fully exploit the information from different scales. Direct fusion of HSI and LiDAR features may not fully exploit their complementarity because of differences in semantic emphasis and scale-dependent responses [38], [39]. Therefore, developing an integrated framework that supports content-aware multiscale feature extraction and adaptive multiscale feature fusion is essential for fully exploiting the complementary characteristics of HSI and LiDAR and improving joint classification performance.

To address the above-mentioned limitations, this study proposes a dual-branch selective multiscale fusion network (DSMF-Net) to jointly model HSI and LiDAR data. The network adaptively captures multiscale spatial structures and performs effective multiscale feature fusion. The primary contributions of this work are summarized as follows.

1) A DSMF-Net is proposed to jointly model and fuse multiscale features from HSI and LiDAR, facilitating deep multimodal interaction.
2) A novel adaptive multiscale convolution module, termed selective multiscale convolution (SMConv), is developed to enable content-driven scale selection via channel-level competition, significantly improving dynamic multiscale feature representation.
3) A modality reconstruction strategy is introduced to regularize multimodal representation learning of multiscale features across heterogeneous modalities.
4) A channel stitching and channel attention (CSCA) module is designed to effectively fuse HSI and LiDAR multiscale features, resulting in more discriminative joint representations.

Beyond performance improvement, this work focuses on how adaptive scale modeling and heterogeneous feature interaction can improve representation learning from hyperspectral and LiDAR data. By combining adaptive receptive-field modeling with structured channel–spatial feature interaction in a unified architecture, the proposed framework provides an effective strategy for multiscale representation learning in complex scenes. These findings contribute to a clearer understanding of how scale adaptation and heterogeneous feature interaction influence representation quality in fine-grained ground-object classification, while also suggesting the potential relevance of the framework for related classification tasks using heterogeneous remote sensing data.

II. Related Works

A. CNN-Based Joint Classification of HSI and LiDAR

The joint classification of HSI and LiDAR data has become an important research topic in remote sensing, due to the strong complementarity between spectral information and 3-D structural characteristics. With the rapid development of deep learning, CNNs have gradually replaced traditional handcrafted-feature-based methods and become the dominant framework for HSI–LiDAR joint classification, due to their powerful hierarchical feature learning capability.

Early CNN-based fusion approaches commonly adopt dual-branch architectures, in which HSI and LiDAR data are processed by two parallel subnetworks to extract modality-specific features, followed by feature fusion at an intermediate or high level. For instance, TBCNN [2] employs two separate CNN branches to learn spectral features from HSI and spatial–structural features from LiDAR, respectively, and concatenates the extracted features for final classification. EndNet [27] further introduces a parallel encoder structure and a fusion-reconstruction mechanism, where multimodal features are jointly learned and constrained through reconstruction loss, significantly enhancing multimodal feature alignment. HRWN [40] extends the dual-branch CNN framework by incorporating a hierarchical random walk module, which models pixel-level affinity relationships to

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exploit spatial dependencies among LiDAR elevation classes. MDL-RS [41] proposes a unified multimodal deep learning framework, combining modality-specific feature extraction networks with flexible fusion strategies to support early, intermediate, and late fusion. More recent studies have explored alternative multimodal fusion strategies within CNN-based HSI–LiDAR classification frameworks. For example, an attention-based HMRF [42] method combines multiscale feature enhancement, a CNN backbone, and posterior-energy-based multimodal modeling to improve spatial consistency and fusion performance. Another representative example is CAB-HL [43], which employs predefined multiscale convolutional branches together with cross-attention-based dual-modality collaboration to strengthen HSI–LiDAR feature interaction.

Recent studies have further explored global dependency modeling for HSI–LiDAR joint classification. Among them, Transformer-based methods have become a mainstream direction in recent studies, as they can capture long-range spatial–spectral dependencies and cross-modal interactions through self-attention mechanisms [44], [45]. More recently, Mamba-based models have also been explored by introducing state-space modeling and selective scanning mechanisms for efficient global feature representation [46]. These studies represent important developments beyond conventional convolution-based fusion frameworks. In contrast, this study focuses on improving multiscale representation and cross-modal fusion within a dual-branch CNN framework, which remains well suited for random-sampling-based training and dense scene-level prediction.

To better preserve spatial details while capturing high-level semantic information, encoder–decoder architectures, particularly U-Net [47] and its variants, have been increasingly adopted in HSI–LiDAR fusion tasks. U-Net employs a symmetric contracting and expansive path with skip connections, allowing low-level spatial details to be directly fused with high-level semantic representations. Its multiscale representation capability primarily arises from this hierarchical encoder–decoder structure, where features are progressively downsampled and aggregated across different resolution levels. In this process, features at different depths correspond to predefined receptive field sizes determined by convolution and pooling operations. Although such hierarchical aggregation enables the integration of information across multiple scales, its scale modeling remains architecturally predetermined and inherently static, lacking an explicit mechanism to adaptively weight scale-specific feature responses according to scene content. Despite this, U-Net-based architectures remain widely adopted in multisource remote sensing fusion.

Overall, despite differences in architectural form, most existing CNN-based methods still rely on fixed receptive fields, predefined kernel branches, or static multiscale configurations. As a result, their ability to cope with large-scale variations in ground objects remains limited. In HSI–LiDAR fusion, this scale rigidity is further exacerbated by heterogeneity, thereby constraining the effectiveness of subsequent feature fusion and unified representation learning.

B. Multiscale CNNs

To address the pronounced scale variations in remote sensing scenes, extensive studies have explored multiscale modeling strategies in deep learning frameworks. According to where multiscale information is introduced, existing approaches can be roughly grouped into three categories.

1) Input-Level Multiscale Strategies: Early methods enhance scale sensitivity by constructing image pyramids or using multiscale neighborhood patches as network inputs [48], [49], [50], [51], [52]. For instance, Farabet et al. [53] represented images using a Laplacian pyramid and applied convolutional networks for multiscale feature extraction. However, such approaches require repeated inference across scales and introduce limited learnable scale adaptation.

2) Operator-Level Multiscale Modeling: The most widely adopted strategy explicitly constructs multiscale receptive fields within convolutional operators. InceptionNets proposed by Szegedy et al. [28], [29], [30] employ parallel convolution kernels of different sizes to capture spatial structures at multiple scales and have been widely extended to remote sensing applications. Following this paradigm, Wang et al. [45] combined 3-D CNNs with inception-style modules for multiscale spatial–spectral feature extraction in HSI–LiDAR fusion. Beyond kernel size variation, dilation-based methods expand receptive fields by adjusting dilation rates. TridentNet [54] introduces multiple branches with different dilation factors, while ASPP [33] aggregates multiscale contextual information via parallel atrous convolutions and has been integrated into residual networks for remote sensing modeling [34]. More recently, GLVMamba [55] combines scale-aware pyramid pooling with global–local state space modeling to enhance multiscale representation learning.

3) Network-Level Multiscale Fusion: Multiscale information can also be integrated through network connectivity design. FPNs [56] fuse multiresolution features via top-down pathways and lateral connections and have become a representative multiscale framework in remote sensing tasks [57]. Methods associated with fully convolutional networks (FCNs) [58] show satisfactory results as well. Zhang et al. [59] further proposed a cascade encoder–decoder network to enhance multiscale feature interaction through hierarchical connectivity.

Despite their effectiveness, these methods generally rely on manually predefined kernel sizes, dilation rates, or fusion topologies, resulting in static scale configurations. Consequently, feature fusion is restricted to fixed or globally weighted combinations over limited candidate scales, lacking content-driven, channelwise adaptive scale selection. In HSI–LiDAR fusion scenarios with large-scale disparities and strong modal heterogeneity, this limitation becomes more pronounced and constrains unified multiscale representation learning.

III. Methods

A. Overview

Motivated by the above observations, we propose a DSMF-Net for joint HSI–LiDAR classification. DSMF-Net

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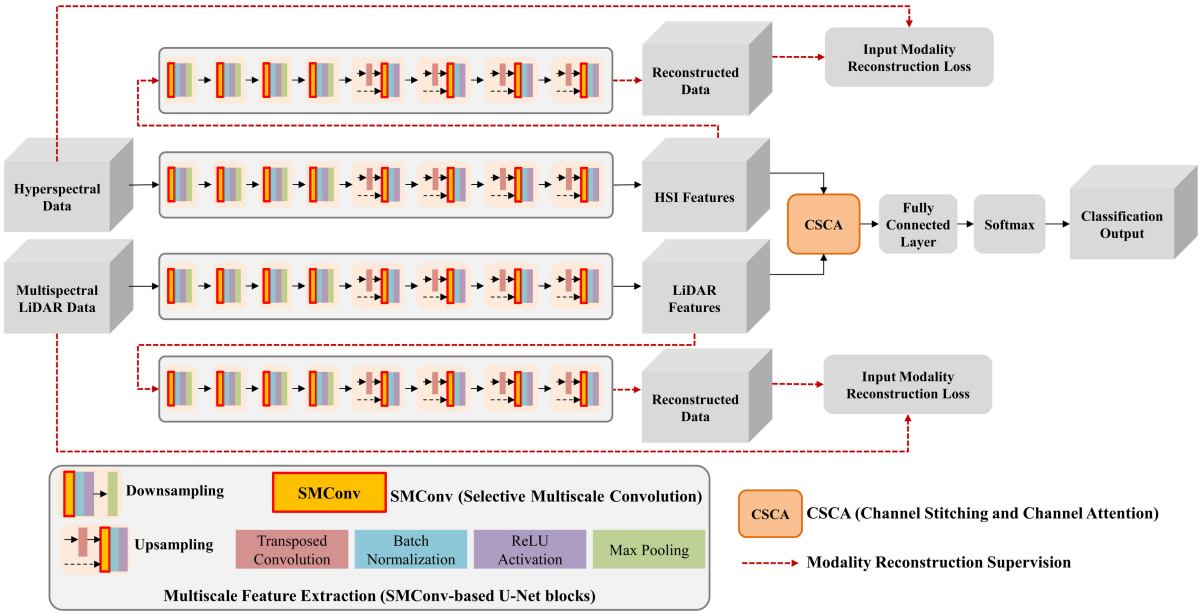


Fig. 1. Overall framework of the proposed DSMF-Net for HSI and LiDAR classification. SMConv is used for adaptive multiscale feature extraction, and CSCA performs multimodal feature fusion. Red dashed arrows indicate modality reconstruction supervision.

is built upon a dual-branch U-Net architecture, where multiscale features are extracted and fused through adaptive receptive-field modeling. The two modality-specific branches focus on learning spectral–spatial characteristics from HSI and structural cues from LiDAR, respectively, enabling effective representation learning. To cope with large-scale variations in real scenes, DSMF-Net introduces a SMConv module, which comprises multiple convolutional branches with distinct receptive fields and an adaptive weighting mechanism for different scales. By dynamically weighting multiscale features extracted by kernels of different sizes according to the input content, SMConv balances local texture preservation with global contextual modeling, thereby improving the scale adaptivity and representation capability of the network.

To further regularize feature learning, a modality reconstruction constraint is incorporated during training to preserve informative representations within each modality. As illustrated in Fig. 1, multilevel features from the two encoders are integrated within the encoder–decoder pathway via skip connections. A CSCA module is employed to enhance the fusion of HSI and LiDAR features by promoting structured channel interaction and adaptively emphasizing informative responses.

Overall, SMConv supports adaptive multiscale representation within each modality, the reconstruction constraint regularizes feature learning, and CSCA promotes effective fusion of HSI and LiDAR features. Through this coordinated design, DSMF-Net achieves adaptive multiscale representation and effective multimodal feature integration within a unified framework.

B. Selective Multiscale Convolution

Ground-object classes in remote sensing imagery often exhibit substantial variation in spatial scale and structural complexity. Convolution kernels with fixed receptive fields are therefore insufficient, as they cannot simultaneously capture fine-grained local patterns and broad contextual structures. To overcome this limitation, we incorporate an SMConv module to replace standard convolutional layers, enabling adaptive receptive-field adjustment and dynamic multiscale feature integration.

As illustrated in Fig. 2, SMConv consists of two convolutional branches with different receptive fields and a channelwise attention unit for adaptive scale selection. To improve computational efficiency, the second branch replaces a standard 5×5 kernel with a dilated 3×3 convolution (dilation $= 2$), which enlarges the effective receptive field without increasing parameter complexity (see Fig. 3). This design enhances multiscale feature extraction at a low computational cost and allows the model to adapt to varying object sizes during pooling and downsampling. The computational efficiency of the proposed design is quantitatively evaluated in terms of model parameters and FLOPs in Section IV.

Given an input feature tensor $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, the SMConv module employs two convolutional branches with different kernel sizes ($K_1 = 3 \times 3$ and $K_2 = 5 \times 5$) to capture both local and wider range spatial responses. The two branches generate feature maps $\mathbf{M}^{(1)} \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{M}^{(2)} \in \mathbb{R}^{H\times W\times C}$, respectively. The outputs of the two branches are then summed elementwise to obtain a fused multiscale feature tensor

$$\mathbf{M} = \mathbf{M}^{(1)} + \mathbf{M}^{(2)}. \tag{1}$$

To retain multiscale information without doubling the parameter count, a group convolution strategy with $G = 2$ is adopted, which effectively reduces computational complexity while maintaining representational capacity. Based on the fused multiscale feature tensor $\mathbf{M}$, a global context vector $\mathbf{z} \in \mathbb{R}^C$ is extracted via global average pooling. In other words,

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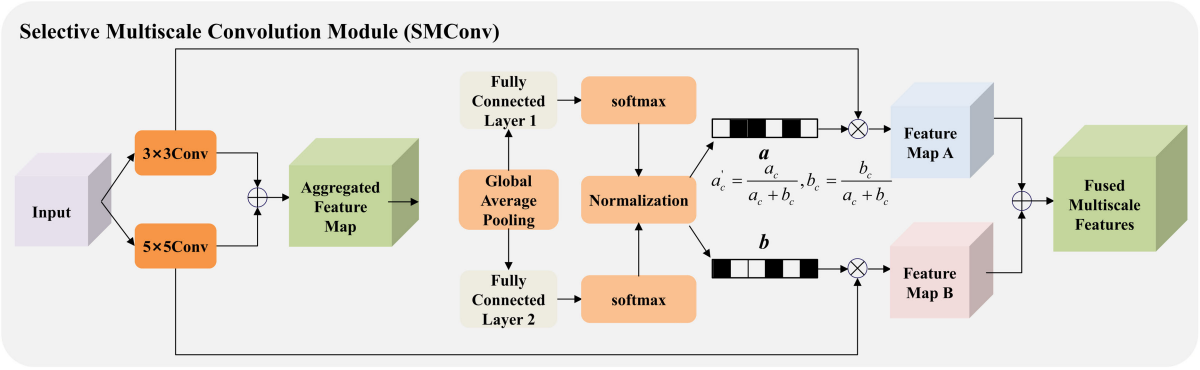


Fig. 2. SMConv module.

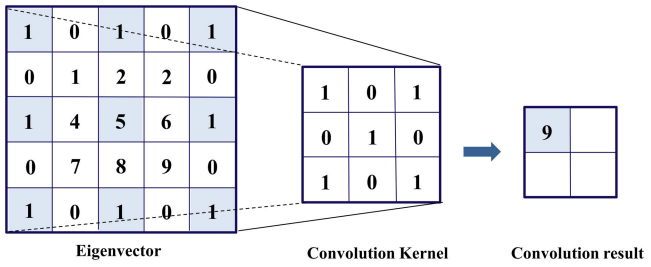


Fig. 3. Illustration of dilated convolution.

the feature map of size H × W is reduced to a single element, with the calculation formula as follows:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} M_{ijc} \tag{2}$$

where z_c denotes the global descriptor of channel c; H and W are the spatial height and width of the feature map, respectively; and M_{ijc} denotes the scalar response at spatial position (i, j) and channel c of the fused feature tensor $\mathbf{M}$.

Building upon the global context vector, a fully connected layer followed by batch normalization (BN) and a ReLU activation is applied to linearly transform the aggregated features, enabling more precise adaptive selection across different scales. Let the resulting vector be denoted as $\mathbf{d}^{(s)} \in \mathbb{R}^C$, where s = 1, 2 corresponds to the responses generated by the two convolutional branches with different kernel sizes. The computation is formulated as follows:

$$\mathbf{d}^{(s)} = f\left(h\left(\mathbf{W}^{(s)}\mathbf{z} + \mathbf{b}^{(s)}\right)\right) \tag{3}$$

where $f(\cdot)$ denotes the activation function (here implemented as ReLU), $h(\cdot)$ represents the BN layer, and $\mathbf{W}^{(s)}$ and $\mathbf{b}^{(s)}$ are the learnable parameters to be estimated during training.

To enable adaptive selection of spatial scales, a softmax operator is applied to compute the probability distribution over feature channels. For each channel, the preliminary importance weights a_c and b_c, corresponding to the two convolutional branches, are computed as follows:

$$a_c = \frac{e^{d_c^{(1)}}}{\sum_{i=1}^{C} e^{d_i^{(1)}}}, \quad b_c = \frac{e^{d_c^{(2)}}}{\sum_{i=1}^{C} e^{d_i^{(2)}}}. \tag{4}$$

Since the two weights jointly determine the contribution of each scale for the same channel, a normalization step is introduced to ensure that their sum equals 1

$$a_c' = \frac{a_c}{a_c + b_c}, \quad b_c' = \frac{b_c}{a_c + b_c}. \tag{5}$$

After obtaining these normalized weights, the multiscale feature representation for channel c is computed as a weighted combination of the outputs from the two branches

$$\mathbf{M}_{:,:,c} = a_c' \cdot \mathbf{M}_{:,:,c}^{(1)} + b_c' \cdot \mathbf{M}_{:,:,c}^{(2)}. \tag{6}$$

Through this process, the module adaptively selects and integrates the most informative scale for each channel, leading to a more discriminative multiscale representation. Although the above formulation is presented for the two-branch case, it can be readily generalized to multiple branches (e.g., three or four kernels with different receptive fields) by extending (1) and (4)–(6).

C. Dual-Branch Encoder–Decoder Network With SMConv

The proposed framework adopts a dual-branch encoder–decoder architecture to fully exploit the complementary characteristics of HSI and LiDAR, as illustrated in Fig. 1. The two branches process HSI and LiDAR independently, enabling the network to capture spectral and structural attributes in a modality-specific manner while maintaining spatial alignment for subsequent fusion. Skip connections are incorporated to preserve fine spatial details and enhance semantic consistency across layers.

To enhance multiscale representation capability, all convolutional layers in the encoder–decoder are replaced with the proposed SMConv module, allowing adaptive receptive-field adjustment at each depth level. The encoder consists of p downsampling blocks, each comprising an SMConv unit followed by BN, ReLU activation, and max pooling. Given inputs $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times C_1}$ and $\mathbf{X}_2 \in \mathbb{R}^{H\times W\times C_2}$, the output of the l-th encoder block is formulated as

$$\begin{cases} \mathbf{Z}_s^{(l)} = f\left(h\left(\text{SMConv}\left(\mathbf{X}_s\right)\right)\right), & l = 1 \\ \mathbf{Z}_s^{(l)} = f\left(h\left(\text{SMConv}\left(\mathbf{Z}_s^{(l-1)}\right)\right)\right), & l = 2, \ldots, p \end{cases} \tag{7}$$

where $s = 1, 2$ denotes the modality branch (HSI or LiDAR), l denotes the block index, and p is the total number of downsampling blocks.

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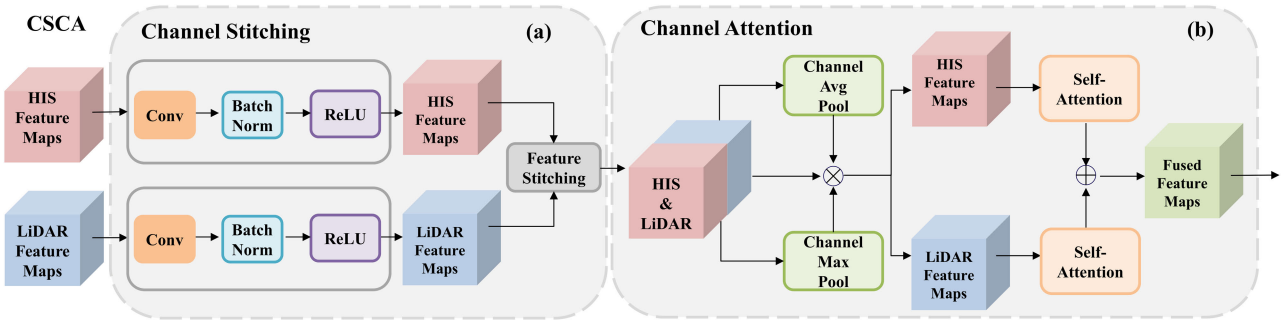


Fig. 4. CSCA module. (a) Channel stitching. (b) Channel attention.

The decoder mirrors the encoder structure to progressively restore spatial resolution while integrating high-level and low-level features. Each upsampling block consists of a transposed convolution, concatenation with the corresponding encoder feature through a skip connection, an SMConv layer, BN, and ReLU activation. Formally, the decoding process is expressed as

$$\begin{cases} \mathrm{M}_s^{(k)} = f\left(h\left(SMConv\left(\mathbf{Z}_s^{(p)}\right)\right)\right), & k=1 \\ \mathrm{M}_s^{(k)} = f\left(h\left(SMConv\left(\mathbf{M}_s^{(k-1)}\right)\right)\right), & k=2,\ldots,p \end{cases} \tag{8}$$

where $\mathrm{M}_s^{(k)}$ denotes the upsampled features at level k. The proposed module is explicitly denoted as $SMConv(\cdot)$.

Inspired by the reverse-decoding strategy introduced in EndNet [27], an auxiliary reconstruction branch is incorporated for each modality to encourage feature preservation and stabilize training. Rather than serving as a generative module, it functions as an auxiliary training regularization mechanism that helps each modality branch retain modality-specific discriminative information during representation learning. The reconstruction process is performed independently within each modality and therefore does not introduce external information or impose artificial equivalence across modalities. This branch is structurally symmetric to the encoder–decoder backbone and is optimized to reconstruct the original input. The reconstruction process is given by

$$\begin{cases} \mathbf{X}_r^{(k)} = g\left(SMConv\left(\mathbf{M}_s\right)\right), & k=1 \\ \mathbf{X}_r^{(k)} = g\left(\mathrm{SMConv}\left(\mathbf{M}_s^{(k-1)}\right)\right), & k=2,\ldots,2p \end{cases} \tag{9}$$

where $g(\cdot)$ denotes the transformation within each decoding layer, including convolution, activation, and normalization.

Minimizing the reconstruction loss forces the modality-specific branches to retain key spectral or structural cues, thereby improving feature robustness and representation stability. Additional ablation results in Appendix A (Table S1) of the Supplementary Material further support the validity of this reconstruction design.

D. Feature Fusion Mechanism

In HSI–LiDAR fusion, multilevel features extracted from different modalities often exhibit heterogeneous statistical distributions and uneven channel activations. Direct concatenation or summation of such features may lead to scale mismatch and dominance of certain channels, resulting in attenuation of complementary information. Particularly in complex urban scenes, discriminative responses corresponding to different object scales may be suppressed during aggregation. Therefore, an effective fusion mechanism should explicitly regulate channelwise interactions and selectively preserve informative multiscale responses. To address these challenges, this study introduces the core concept of the channel switching and spatial attention (CSSA) module [60], which provides a lightweight, parameter-free spatial attention mechanism. Specifically, CSSA generates spatial attention weights by applying channel average pooling (CAP) and channel max pooling (CMP) to concatenated multimodal features, enabling feature-level weighted fusion that highlights informative responses while suppressing redundant information.

However, CSSA primarily focuses on spatial reweighting and does not explicitly model channelwise interactions. To further enhance the utilization of multiscale features, we extend CSSA by introducing a CSCA module (see Fig. 4), which incorporates channel-level interaction and self-attention mechanisms to strengthen discriminative feature fusion between HSI and LiDAR.

Specifically, to accommodate the heterogeneous channel structures of the two modalities, we first perform channel stitching and reconstruction using a 1×1 convolution followed by BN and a ReLU activation

$$\mathbf{M}_s' = f\left(h\left(w_{1\times1}\left(\mathbf{M}_s\right)\right)\right) \tag{10}$$

where $w_{1\times1}(\cdot)$ denotes the 1×1 convolution operator.

This operation linearly transforms and compresses the channel representations of each modality without altering their spatial resolution. The reconstructed features from the two modalities are then concatenated along the channel dimension to form the fused feature map $\mathbf{M}_{cat}\in\mathbb{R}^{H\times W\times 2C}$

$$\mathbf{M}_{cat} = concat\left(\mathbf{M}_1', \mathbf{M}_2'\right). \tag{11}$$

A simple concatenation of features from the two modalities fails to explicitly model the interactions between passive and active remote sensing data, thereby limiting the effectiveness of multimodal feature fusion. Moreover, as the number of channels increases, the computational cost of channel attention (CA) also grows significantly. To enable efficient fusion while controlling model complexity, CAP and CMP are introduced within the channel stitching attention module. Specifically, CAP captures global channel responses via mean aggregation,

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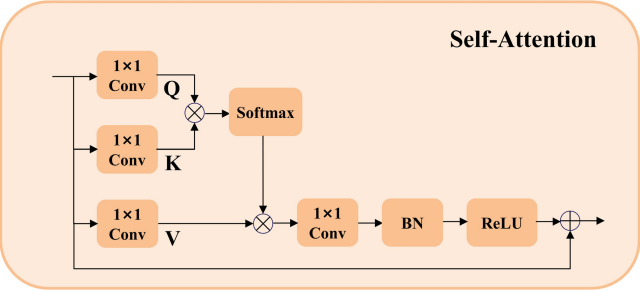


Fig. 5. Structure of the self-attention mechanism.

whereas CMP emphasizes salient features using a max operator [56]. The calculation formula is as follows:

$$\mathrm{CAP}(\mathbf{M})_{ij} = \frac{1}{C}\sum_{c=1}^{C} M_{ijc} \tag{12}$$

$$\mathrm{CMP}(\mathbf{M})_{ij} = \max_{1\le c\le C} M_{ijc} \tag{13}$$

where $(\mathbf{M})_{ij}$ denotes the channel vector at spatial position (i, j), and C represents the number of feature channels. This operation effectively compresses information without increasing model complexity. The outputs of CAP and CMP are then used as the inputs to the CA mechanism. The concatenated feature tensor $\mathbf{M}_{cat} \in \mathbb{R}^{H\times W\times 2C}$ is first processed by CAP and CMP to generate two spatial weight maps, $\mathbf{W}_{ave} \in \mathbb{R}^{H\times W\times 1}$ and $\mathbf{W}_{max} \in \mathbb{R}^{H\times W\times 1}$, respectively. Subsequently, weighted feature fusion is performed through matrix multiplication among the three feature maps, as expressed as follows:

$$\mathbf{M}_{cat}^{w} = \sigma(\mathbf{M}_{cat}) \otimes \sigma(\mathbf{W}_{ave}) \otimes \sigma(\mathbf{W}_{max}) \tag{14}$$

where $\otimes$ denotes the matrix multiplication and $\sigma(\cdot)$ represents the Sigmoid activation function applied prior to multiplication. After obtaining the weighted fused feature map, feature splitting is performed to produce modality-specific weighted subfeature maps for HSI and LiDAR data, as shown in the following equation:

$$\mathbf{M}_{H}^{w}, \mathbf{M}_{L}^{w} = \mathrm{Split}\left(\mathrm{M}_{cat}^{w}\right) \tag{15}$$

where $\mathbf{M}_{H}^{w}, \mathbf{M}_{L}^{w} \in \mathbb{R}^{H\times W\times C}$.

In addition, to further capture the interchannel dependencies between the HSI and LiDAR features, a Transformer-style self-attention mechanism is introduced (see Fig. 5). Specifically, the input feature map is first projected into query, key, and value representations through three independent 1×1 convolutions, followed by the corresponding dimensional transformations. The attention weights $\mathbf{W}_a \in \mathbb{R}^{HW\times C\times C}$ are obtained by multiplying $\mathbf{Q} \in \mathbb{R}^{HW\times C\times 1}$ with $\mathbf{K} \in \mathbb{R}^{HW\times 1\times C}$, followed by Softmax normalization:

$$\mathbf{W}_a = \mathrm{Softmax}\left(\left(w_{1,q}(\mathbf{M}_{cat}) \otimes w_{1,k}(\mathbf{M}_{cat})\right)\right). \tag{16}$$

Matrix multiplication is then performed between the attention weights and $\mathbf{V} \in \mathbb{R}^{HW\times 1\times C}$ to obtain the weighted feature map. Finally, add the input vector to obtain the fused feature map $\mathbf{M}_{fused}^{w} \in \mathbb{R}^{H\times W\times C}$. The specific calculation process is as follows:

$$\mathbf{M}_{fused}^{w} = f\left(h\left(\sigma\left(w_{1,a} \otimes \left(w_{1,v}(\mathbf{M}_{cat})\right)\right)\right)\right) + \mathbf{M}_{cat} \tag{17}$$

where $w_1(\cdot)$ denotes the 1×1 convolution operator. The final fusion result is expressed as

$$\mathbf{M}_{fused} = \frac{\mathbf{M}_{H,fused}^{w} + \mathbf{M}_{L,fused}^{w}}{2} \tag{18}$$

where $\mathbf{M}_{H,fused}^{w}$ and $\mathbf{M}_{L,fused}^{w}$ denote the fusion feature submaps of hyperspectral image data and LiDAR data computed through the CA layer, respectively.

IV. Experiments and Results

A. Dataset

This study employs the publicly available IEEE Data Fusion Contest 2018 (DFC2018 Houston) dataset, collected by the National Center for Airborne Laser Mapping (NCALM) in February 2017. As a widely used benchmark for HSI–LiDAR classification, the dataset covers the University of Houston campus and the surrounding urban area, and contains complex urban scenes with substantial spectral, structural, and spatial variability. It provides both HSI and multispectral LiDAR (MS-LiDAR) data.

1) Hyperspectral Imagery: The hyperspectral image was acquired by an airborne imaging spectrometer and contains 48 spectral bands covering the 380–1050-nm range with an approximate spectral resolution of 10 nm and a spatial resolution of 2.5 m. Due to its high spectral resolution, the HSI enables fine discrimination of ground-object types based on subtle spectral variations. However, the relatively coarse spatial resolution can lead to spectral mixing in heterogeneous urban environments.

2) Multispectral LiDAR: The MS-LiDAR data were acquired using the Titan MS-LiDAR system, which operates three active imaging channels with center wavelengths at 532 nm (green), 1064 nm (near-infrared), and 1550 nm (shortwave infrared). The system simultaneously provides surface elevation information (DSM) and multiwavelength intensity measurements, capturing both structural and spectral characteristics of urban objects. For multimodal integration, the HSI and LiDAR-derived DSM data were co-registered and resampled to a common spatial resolution within the same coordinate system.

3) Ground Truth: The dataset includes ground-truth labels for 20 representative ground-object classes, spanning buildings, roads, trees, water bodies, grasses, and other urban targets exhibiting substantial spectral and structural diversity. In this study, the full hyperspectral image, the rasterized and preprocessed LiDAR-derived DSM, and the corresponding ground-truth annotations were used for model training and evaluation. The diversity of classes and scene complexity make this dataset suitable for evaluating the ability of the proposed framework to model multiscale spatial patterns and heterogeneous features in fine-grained classification.

B. Experimental Setup

To evaluate the effectiveness of the proposed self-adaptive receptive-field multiscale convolution module (SMConv), a series of experiments were conducted on the DFC2018 Houston dataset. All experiments were implemented

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TABLE I
CLASSIFICATION METRICS ACROSS DIFFERENT MODELS

No.	RF [61]	TBCNN [2]	EndNet [27]	HRWN [40]	MDL-RS [41]	Ours
C1	45.05 ± 1.17	47.56 ± 4.37	77.08 ± 0.93	78.19 ± 0.33	79.48 ± 1.21	**87.86 ± 0.89**
C2	55.37 ± 0.51	40.73 ± 0.56	87.83 ± 0.74	88.39 ± 0.42	80.90 ± 0.79	**94.96 ± 0.09**
C3	26.21 ± 6.36	18.71 ± 6.11	78.95 ± 1.00	65.06 ± 3.20	44.19 ± 3.86	**92.47 ± 1.46**
C4	17.30 ± 2.46	38.18 ± 6.15	84.52 ± 1.05	78.74 ± 3.68	75.83 ± 1.39	**95.99 ± 0.07**
C5	4.15 ± 0.57	46.96 ± 0.68	86.63 ± 0.72	78.07 ± 0.70	76.47 ± 1.75	**94.88 ± 0.13**
C6	51.35 ± 0.93	52.56 ± 0.73	97.53 ± 0.15	82.84 ± 1.61	82.14 ± 2.04	**97.76 ± 1.10**
C7	71.99 ± 15.67	62.41 ± 0.71	**86.56 ± 1.31**	73.59 ± 2.48	37.59 ± 4.09	75.47 ± 2.53
C8	57.00 ± 0.72	70.20 ± 0.89	90.30 ± 0.43	89.57 ± 0.07	88.71 ± 0.22	**96.48 ± 0.10**
C9	78.42 ± 0.19	91.29 ± 0.28	97.79 ± 0.63	97.63 ± 0.13	97.05 ± 0.10	**98.65 ± 0.14**
C10	10.27 ± 0.32	45.24 ± 2.02	71.46 ± 2.88	64.86 ± 2.08	68.72 ± 1.34	**90.79 ± 0.24**
C11	2.91 ± 0.16	49.88 ± 0.48	68.03 ± 0.56	46.22 ± 1.86	68.46 ± 0.80	**83.56 ± 0.31**
C12	1.44 ± 0.25	21.42 ± 1.64	50.68 ± 0.29	43.59 ± 0.59	46.99 ± 1.30	**75.33 ± 0.98**
C13	29.98 ± 0.46	80.85 ± 0.67	92.57 ± 0.14	89.24 ± 0.17	90.03 ± 0.42	**97.55 ± 0.03**
C14	50.19 ± 0.71	20.53 ± 2.67	93.62 ± 0.45	92.97 ± 0.62	90.59 ± 1.56	**98.42 ± 0.10**
C15	56.61 ± 3.87	49.82 ± 0.74	92.77 ± 0.95	93.83 ± 0.04	90.35 ± 0.36	**95.78 ± 0.31**
C16	23.59 ± 1.05	50.71 ± 2.60	88.26 ± 0.84	73.87 ± 1.21	78.27 ± 1.23	**95.93 ± 0.23**
C17	4.77 ± 0.94	59.63 ± 0.42	86.03 ± 0.77	73.76 ± 2.97	89.95 ± 0.70	**98.30 ± 0.28**
C18	1.87 ± 0.86	15.79 ± 2.48	62.98 ± 0.98	48.05 ± 1.82	42.02 ± 1.03	**80.23 ± 0.83**
C19	29.19 ± 1.73	73.09 ± 0.38	87.18 ± 0.47	84.40 ± 0.99	92.54 ± 0.64	**96.51 ± 0.37**
C20	76.09 ± 1.17	90.73 ± 0.17	97.03 ± 0.54	94.50 ± 0.08	96.66 ± 1.12	**96.15 ± 0.42**
OA	80.59 ± 0.06	85.77 ± 0.09	93.92 ± 0.12	94.30 ± 0.07	92.53 ± 0.25	**97.16 ± 0.10**
AA	67.62 ± 0.56	75.92 ± 0.24	91.95 ± 1.09	88.61 ± 1.41	88.03 ± 0.76	**96.09 ± 0.71**
Kappa	0.586 ± 0.013	0.726 ± 0.060	0.889 ± 0.029	0.892 ± 0.005	0.863 ± 0.011	**0.948 ± 0.002**
mIoU	33.05 ± 0.41	45.41 ± 0.14	74.14 ± 0.15	74.85 ± 0.56	67.32 ± 1.40	**86.38 ± 0.47**

Note: Bold text indicates the best result for each metric, while underlined text indicates the second-best result.

within a dual-branch U-Net architecture, where the HSI and MS-LiDAR data were processed by two separate encoder branches to preserve the spectral and structural consistency of the active–passive modalities. All convolutional units in the network were replaced with the proposed SMConv module to enable adaptive receptive-field adjustment and multiscale feature extraction.

To ensure spatial consistency between the two modalities, the HSI and MS-LiDAR DSM were first co-registered and resampled to a common spatial resolution. During training, 128×128 image patches were randomly cropped from the full scene and normalized on a per-channel basis. Approximately 10% of the labeled pixels were randomly selected as the training set, while the remaining 90% were used for testing. A binary mask was maintained to record the training-pixel locations, and the loss was computed only on pixels with mask = 1 to prevent unintended gradient updates from test samples.

The network was implemented in PyTorch and optimized using the Adam optimizer, with an initial learning rate of 1×10^{-3}, a weight decay of 1×10^{-4}, and a batch size of 8. Training was performed for 200 epochs with early stopping on the validation set to mitigate overfitting. Cross-entropy loss was adopted for supervising the pixelwise classification. To improve the robustness of model training and reduce overfitting, basic data augmentation strategies were applied during training, including random horizontal and vertical flipping, random rotations ($\pm 90°$), and Gaussian noise injection into the hyperspectral channels. Each augmentation operation was applied with a probability of 0.5. All comparative experiments were conducted under identical input configurations, optimization settings, and training protocols to ensure fair and reproducible results.

All experiments were performed on an NVIDIA GeForce RTX 3090. Each model was trained from randomly initialized weights, and the final results were reported as the average over five independent runs to reduce the impact of stochastic variations.

C. Performance Comparison

To evaluate the effectiveness of the proposed DSMF-Net, several representative CNN-based HSI–LiDAR joint classification methods are selected as comparison baselines, including TBCNN [2], EndNet [27], HRWN [40], and MDL-RS [41]. A detailed description of these methods can be found in Section II-A. In addition, random forest (RF) is included as a traditional machine-learning baseline [61]. All models were implemented using publicly available code and trained under the same data split, input size, and optimization settings to ensure fair comparison. Four widely used metrics were employed to quantitatively assess classification performance: overall accuracy (OA), average accuracy (AA), the Kappa coefficient, and mean intersection over union (mIoU). These metrics jointly evaluate overall correctness, class-

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wise balance, and spatial agreement between predictions and ground truth.

Table I presents the quantitative comparison between the proposed model and five classification methods, including RF, TBCNN, EndNet, HRWN, MDL-RS. The results are reported as mean ± standard deviation over five independent runs. The results demonstrate that our method achieves the best performance across all evaluation metrics on the DFC2018 Houston dataset.

The proposed model reaches an OA of 97.16%, outperforming the strongest deep-learning baseline (HRWN, 94.30%) by 2.86%. The AA increases to 96.09%, exceeding EndNet by 4.14%. In addition, the proposed approach yields a Kappa coefficient of 0.948 and an mIoU of 86.38%, indicating superior classification consistency and spatial reliability. In terms of the overall metrics, the proposed method shows relatively small standard deviations across repeated runs, indicating stable performance under different random initializations. In contrast, RF yields the lowest performance among all compared methods, achieving an OA of 80.59% and an mIoU of 33.05%. This is mainly attributed to its limited ability to model complex spectral–spatial dependencies and multiscale contextual structures in HSI–LiDAR data, resulting in weaker classwise balance and spatial consistency compared with deep-learning-based approaches.

From the perspective of class-level accuracy, the proposed method obtains the highest accuracy in 19 out of 20 categories. Particularly notable improvements are observed in challenging classes such as C3 (grass-man-made), C4 (evergreen trees), C8 (residential), C15 (railway), and C17 (unpaved parking), all exceeding 95% accuracy. These results highlight the robustness of the proposed model in handling heterogeneous spatial structures and subtle spectral variations.

Compared with traditional models (e.g., RF and TBCNN), which struggle with spectral mixing and ambiguous boundaries, our method benefits from the adaptive receptive-field capability of SMConv, enabling more discriminative multiscale representation learning. Furthermore, the CSCA-based feature fusion enhances interchannel interactions between hyperspectral and LiDAR modalities, improving both accuracy and spatial consistency. As illustrated in Fig. 6, the proposed fusion strategy produces classification maps that better align with real ground-object distributions, with fewer noise artifacts and sharper boundaries. Overall, the experimental results clearly demonstrate the superiority of the proposed model in HSI–LiDAR fusion classification tasks.

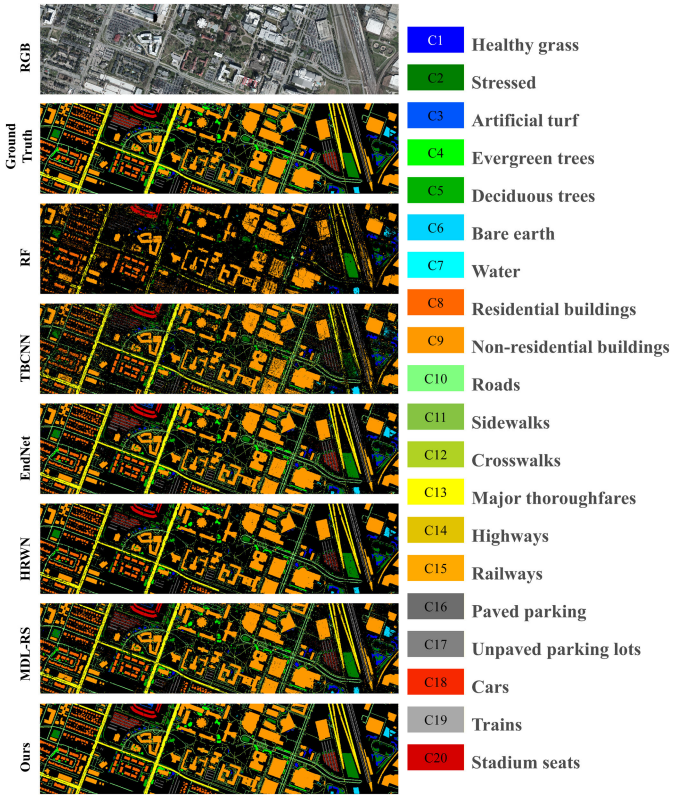


Fig. 6. Classification results diagram for dataset visualization across different models.

V. Discussion

A. Ablation Study

To further quantify the contribution of the proposed SMConv module, an ablation study was conducted under the same dual-branch U-Net framework equipped with the CSCA-based feature fusion module. Specifically, we compare a baseline model that employs standard convolutions with an enhanced model in which all convolutional layers are replaced by SMConv. The results are reported in Table II.

As shown in Table II, incorporating SMConv consistently improves performance across all evaluation metrics. The OA increases from 96.82% to 97.16%, while the AA improves from 95.94% to 96.09%. The Kappa coefficient rises from 0.9424 to 0.9484, and the mIoU increases from 84.72% to 86.38%. Although the OA improvement brought by SMConv is relatively modest, this comparison is conducted on top of an already strong U-Net + CSCA baseline. The improvement is more evident in mIoU and several challenging categories, such as crosswalk, vehicles, and artificial turf, indicating that SMConv is particularly beneficial for small-scale or structurally complex categories, where adaptive multiscale feature representation is more critical for accurate discrimination. Meanwhile, the additional computational cost remains acceptable, with parameters increasing from 30.69 M to 34.22 M and FLOPs from 34.10 G to 36.24 G, suggesting a reasonable tradeoff between accuracy improvement and computational overhead.

At the class level, the SMConv-enhanced model achieves higher accuracies in most ground-object categories, with particularly notable improvements in fine-grained or structurally complex classes such as C3 (artificial turf), C12 (crosswalk), and C18 (vehicles), where accuracy gains reach 4%–6%. This demonstrates that SMConv effectively captures multiscale spatial patterns through adaptive receptive-field adjustment and enhances discrimination in regions with complex textures or ambiguous boundaries. However, the accuracy for Water (C7) decreases from 87.97% to 75.47%, likely because water regions are usually homogeneous and have relatively simple boundaries, so the benefit of multiscale feature enhancement is limited, while extra contextual information may increase boundary confusion. Moreover, water occupies a relatively

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TABLE II
ABLATION EXPERIMENTS ON THE HOUSTON2018 DATASET FOR THE PROPOSED METHOD

No.	Class Name	U-Net + CSCA	U-Net + CSCA + SMConv
C1	Healthy Grass	**90.76**	87.86
C2	Stressed Grass	94.31	**94.96**
C3	Artificial Turf / Grass-Manmade	88.85	**92.47**
C4	Evergreen Trees	**96.03**	95.99
C5	Deciduous Trees	94.70	**94.88**
C6	Bare Soil	97.67	**97.76**
C7	Water	**87.97**	75.47
C8	Residential	94.96	**96.48**
C9	Non-residential	**98.94**	98.65
C10	Road	89.64	**90.79**
C11	Sidewalk	82.37	**83.56**
C12	Crosswalk	69.63	**75.33**
C13	Main Thoroughfare / Main Road	96.17	**97.55**
C14	Highway	98.32	**98.42**
C15	Railway	**96.69**	95.78
C16	Parking Lot 1 (Paved)	95.55	**95.93**
C17	Parking Lot 2 (Unpaved)	96.59	**98.30**
C18	Vehicles	74.28	**80.23**
C19	Train	96.48	**96.51**
C20	Stadium Seats	**97.89**	96.15
OA	Overall Accuracy	96.82	**97.16**
AA	Average Accuracy	95.94	**96.09**
Kappa	Kappa Coefficient	0.9424	**0.9484**
mIoU	Mean IoU	84.72	**86.38**
Params (M)	Parameters (Million)	30.69	**34.22**
FLOPs (G)	FLOPs (Giga)	34.10	**36.24**

Note: Bold text indicates the best result for each metric.

small proportion of the scene and is spatially concentrated in only a few localized regions. Under the random patch-sampling strategy, this may reduce its effective sampling frequency during training and lead to insufficient representation of water-specific patterns.

Qualitative comparisons in Fig. 6 further support these observations. With SMConv, the predicted classification maps exhibit fewer noisy pixels, cleaner object boundaries, and overall better alignment with the reference ground-object distribution. These results confirm that SMConv substantially strengthens the feature representation capacity and spatial discriminability of the network, providing a robust and effective enhancement for HSI–LiDAR fusion classification.

B. Effect of Kernel Scale in SMConv

To further examine the influence of kernel scale within the dual-branch multiscale feature extraction structure, the first branch was fixed to a 3×3 convolution, while the second branch was configured with different kernel sizes (3×3, 5×5, and 7×7). The corresponding results are reported in Table III.

As shown in the table, the configuration using a 7×7 kernel in the second branch achieves the best performance

TABLE III
MODEL ACCURACY WHEN USING DIFFERENT-SCALE CONVOLUTIONAL KERNELS IN THE SECOND MULTISCALE FEATURE SELECTION MODULE

Convolutional kernels scale	OA	AA	Kappa	mIoU
3×3	96.42	93.28	0.9347	80.58
5×5	<u>97.16</u>	**96.09**	<u>0.9484</u>	<u>86.38</u>
7×7	**97.28**	<u>95.78</u>	**0.9505**	**86.50**

Note: Bold text indicates the best result among all metrics, while underlined text indicates the next best result.

(OA = 97.28%, AA = 95.78%, Kappa = 0.9505, and mIoU = 86.50%), followed by the 5×5 setting, both of which clearly outperform the equal-scale 3×3–3×3 configuration. This demonstrates that introducing scale diversity in the dual-branch structure enriches feature representation and enhances the model's ability to perceive ground objects of varying spatial sizes.

When both branches adopt the same kernel size (3×3), the identical receptive fields lead to high feature redundancy, and the subsequent multiscale fusion introduces interfering information, resulting in reduced accuracy. In contrast, larger kernels (5×5 and 7×7) expand the receptive field and enable better modeling of broader spatial patterns while preserving local detail, thereby improving classification performance in complex scenes. These findings further validate the importance of multiscale information in HSI–LiDAR fusion tasks and confirm the effectiveness and soundness of the proposed multiscale feature extraction module.

C. Multibranch Combination Strategies

To further examine the structural flexibility of the proposed multiscale module, we extend the dual-branch design to a multibranch configuration by combining multiple kernel sizes (3×3, 5×5, and 7×7). Different kernel combinations were evaluated, as summarized in Table IV. In the table, the rows marked "SMConv" indicate that the adaptive fusion mechanism of SMConv was applied across the selected branches; otherwise, the outputs of the individual kernels were simply summed.

As shown in Table IV, SMConv enables more effective exploitation of multiscale information than direct summation, consistently yielding higher classification accuracy. Among all configurations, the combination of 3×3 and 7×7 kernels within SMConv achieves the best performance (OA = 97.28%, AA = 95.78%, Kappa = 0.9505, and mIoU = 86.50%), indicating that pairing complementary receptive fields provides stronger feature diversity than using adjacent kernel sizes.

In addition to the dual-branch configurations, we also evaluated three-branch combinations ($3 \times 3 + 5 \times 5 + 7 \times 7$). Although the three-branch SMConv structure produces slightly lower accuracy than the optimal dual-branch setting, it still improves upon the non-SMConv baseline (OA increasing from 96.82% to 96.96% and Kappa from 0.9424 to 0.9450). This suggests that incorporating additional scales is beneficial up to a point, but excessive scale diversity may introduce redundant information and limit performance gains.

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TABLE IV
MODEL ACCURACY WITH DIFFERENT CONVOLUTION KERNEL PAIRING STRATEGIES

SMConv	K=3×3	K=5×5	K=7×7	OA	AA	Kappa	mIoU
	√	√		96.85	96.19	0.9429	85.52
√	√	√		97.16	96.09	0.9484	86.38
	√		√	97.06	95.71	0.9466	86.11
√	√		√	**97.28**	95.78	**0.9505**	**86.50**
		√	√	97.13	95.83	0.9478	86.43
√		√	√	97.18	**96.35**	0.9488	86.33
	√	√	√	96.90	95.60	0.9436	85.33
√	√	√	√	96.96	95.08	0.9450	83.83

Note: Bold text indicates the best result among all metrics, while underlined text indicates the next best result.

TABLE V
CLASSIFICATION RESULTS AFTER EMBEDDING SMCONV INTO DIFFERENT CNN-BASED MODELS

Method	OA	AA	Kappa	mIoU
TBCNN [2]	85.77	75.92	0.7262	45.41
TBCNN [2]+SMConv	**93.05**	**85.77**	**0.8689**	**67.39**
EndNet [27]	93.92	**91.95**	0.8887	74.14
EndNet [27]+SMConv	**95.65**	91.26	**0.9201**	**77.37**
HRWN [40]	94.30	88.61	0.8920	74.85
HRWN [40]+SMConv	**95.31**	**90.39**	**0.9124**	**77.81**
MDL-RS [41]	92.53	88.03	0.8630	67.32
MDL-RS [41]+SMConv	**94.39**	**88.44**	**0.8961**	**72.25**

Note: Bold text indicates the best result for each metric.

Overall, the results confirm that multiscale information, regardless of the specific kernel pairing, provides clear benefits for HSI–LiDAR classification. The experiments further demonstrate the effectiveness and design soundness of the proposed SMConv module in adaptively integrating multibranch features.

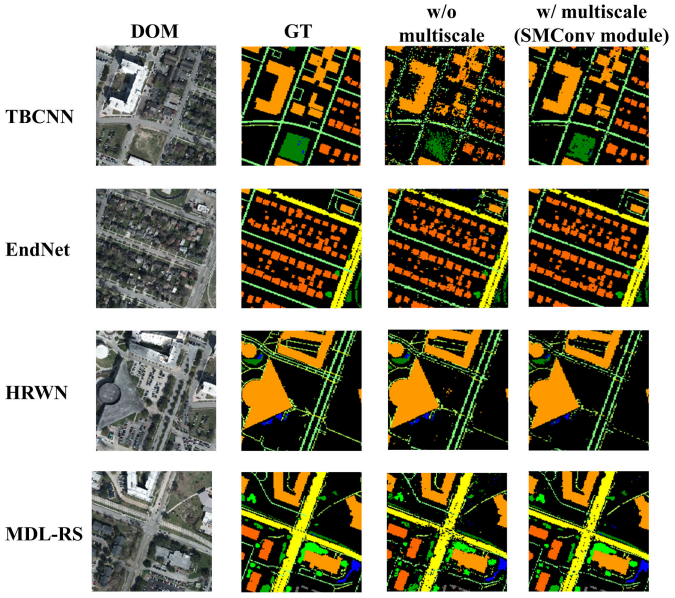


Fig. 7. Impact of multiscale information on classification results of different methods (DOM: digital orthophoto map; GT: ground truth).

D. Adaptability of the Proposed Module Across Different CNN Backbones

1) Quantitative Analysis: To further assess the adaptability and backbone-level robustness of the proposed SMConv module, we integrated it into several representative backbone networks, including TBCNN, EndNet, HRWN, and MDL-RS, and conducted experiments on the DFC2018 Houston dataset. The results are summarized in Table V.

As shown in the table, incorporating SMConv consistently improves the performance of all backbone models. EndNet, HRWN, and MDL-RS all benefit from the introduction of SMConv, with OA increasing from 93.92% to 95.65%, 94.30% to 95.31%, and 92.53% to 94.39%, respectively. For HRWN and MDL-RS, AA also improves from 88.61% to 90.39% and from 88.03% to 88.44%, respectively, while EndNet shows a slight decrease in AA from 91.95% to 91.26%. The Kappa coefficient and mIoU are consistently improved across all models, indicating that SMConv effectively enhances multiscale feature representation and improves discrimination in complex scenes. Notably, TBCNN shows the largest improvement after introducing SMConv, with OA increasing by 7.28 percentage points (from 85.77% to 93.05%). This confirms the strong adaptability of SMConv across different architectures.

Overall, these results validate the robustness and applicability of SMConv. Regardless of whether it is integrated into lightweight or deep architectures, the module consistently delivers performance gains, highlighting its potential for broader use in HSI–LiDAR fusion and other multimodal remote sensing tasks.

2) Visual Comparison and Analysis: As illustrated in Fig. 7, incorporating the SMConv module noticeably improves the visual quality of classification results across all four backbone networks. After introducing multiscale features, the predicted maps exhibit substantially fewer noisy pixels, and the delineation of objects at different spatial scales becomes clearer. Large-scale classes (e.g., roads and nonresidential buildings) show sharper boundaries and reduced misclassification, while small-scale objects (e.g., residential structures and trees) also benefit from enhanced local detail representation.

These qualitative improvements further validate the effectiveness of SMConv in capturing multiscale spatial patterns. By adaptively adjusting the receptive field, SMConv enables the network to better discriminate structurally complex

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TABLE VI
EFFECT OF THE SMCONV MODULE ON MODEL PARAMETERS AND COMPUTATIONAL LOAD

Model	Parameters (M)	FLOPs (G)
TBCNN [2]	**0.35**	**11.50**
TBCNN [2]+SMConv (standard convolution)	0.71	21.99
TBCNN [2]+SMConv (grouped convolution)	<u>0.40</u>	<u>12.03</u>
EndNet [27]	**0.80**	**26.19**
EndNet [27] +SMConv (standard convolution)	1.67	52.06
EndNet [27]+SMConv (grouped convolution)	<u>0.89</u>	<u>26.69</u>
HRWN [40]	**1.43**	**46.76**
HRWN [40]+SMConv (standard convolution)	2.43	75.66
HRWN [40]+SMConv (grouped convolution)	<u>1.73</u>	<u>52.60</u>
MDL-RS [41]	**0.58**	**49.02**
MDL-RS [41]+SMConv (standard convolution)	1.23	97.88
MDL-RS [41]+SMConv (grouped convolution)	<u>0.67</u>	<u>50.46</u>

Note: Bold text indicates the best result among all metrics, while underlined text indicates the next best result.

or spectrally mixed regions. The consistent enhancements observed across different CNN-based architectures demonstrate the applicability and backbone-level robustness of the proposed module.

E. Computational Efficiency and Complexity Analysis

Table VI reports the changes in parameter count and computational cost (floating point operations, FLOPs) after integrating the proposed SMConv module into four representative backbone networks: TBCNN, EndNet, HRWN, and MDL-RS. For each model, all fixed-size convolution layers were replaced with SMConv while keeping the overall architecture unchanged. Both nongrouped and grouped versions of SMConv were evaluated. The number of FLOPs is calculated based on the input tensors with sizes of $48 \times 128 \times 128$ and $3 \times 128 \times 128$, where 48 and 3 represent the number of channels for hyperspectral images and MS-LiDAR data, respectively.

As shown in the table, directly inserting the nongrouped SMConv substantially increases the number of parameters and FLOPs due to the use of multiple convolution kernels of different sizes. In contrast, the grouped-convolution variant achieves multiscale feature extraction with only marginal increases in computational cost, indicating that grouped SMConv provides a more efficient and lightweight integration strategy.

VI. CONCLUSION

In this article, we proposed a multiscale convolution module with adaptive receptive fields (SMConv) and developed a dual-branch framework for hyperspectral and MS-LiDAR feature fusion. By dynamically adjusting its effective receptive field according to spatial–structural variations, SMConv enables more flexible and discriminative multiscale feature extraction, thereby enhancing the representation of heterogeneous ground-object classes in multimodal remote sensing data.

Extensive experiments on the DFC2018 Houston dataset demonstrate that the proposed method achieves superior performance across multiple evaluation metrics, particularly in complex and fine-grained ground-object categories. Ablation studies further validate the design rationality of SMConv by revealing the impact of kernel scale, branch combinations, and multiscale information integration on classification accuracy. Additional experiments across different CNN-based backbone networks show that SMConv consistently improves the performance of various backbone networks while maintaining favorable parameter and computational efficiency through grouped convolution, highlighting its applicability and architectural adaptability.

Although this study validates the proposed framework on the DFC2018 Houston dataset, further cross-dataset evaluation on additional HSI–LiDAR datasets remains necessary to assess its transferability under different sensor configurations, class distributions, and spatial resolutions. Future work will focus on extending the framework to multitemporal and multiregion datasets to improve robustness under dynamic environmental conditions. Further investigation will also be directed toward the interpretability of the adaptive receptive-field mechanism and its role in scale selection. We will also explore lightweight variants of SMConv for large-scale applications.

REFERENCES

[1] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[2] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018, doi: 10.1109/TGRS.2017.2756851.

[3] L. Gao, G. Chai, and X. Zhang, "Above-ground biomass estimation of plantation with different tree species using airborne LiDAR and hyperspectral data," *Remote Sens.*, vol. 14, no. 11, p. 2568, May 2022, doi: 10.3390/rs14112568.

[4] O. Reisi Gahrouei, J.-F. Côté, P. Bournival, P. Giguère, and M. Béland, "Comparison of deep and machine learning approaches for Quebec tree species classification using a combination of multispectral and LiDAR data," *Can. J. Remote Sens.*, vol. 50, no. 1, Dec. 2024, Art. no. 2359433, doi: 10.1080/07038992.2024.2359433.

[5] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020, doi: 10.1109/TGRS.2020.2969024.

[6] R. Luo et al., "Fusion of hyperspectral and LiDAR data for classification of cloud-shadow mixed remote sensed scene," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3768–3781, Aug. 2017, doi: 10.1109/JSTARS.2017.2684085.

[7] M. Dalponte, L. Bruzzone, and D. Gianelle, "Fusion of hyperspectral and LiDAR remote sensing data for classification of complex forest areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1416–1427, May 2008, doi: 10.1109/TGRS.2008.916480.

[8] A. Wang et al., "Integration of LiDAR and hyperspectral imagery for tree species identification at the individual tree level," *Photogramm. Rec.*, vol. 40, no. 190, p. 70007, Apr. 2025, doi: 10.1111/phor.70007.

[9] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522416.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:03:12 UTC from IEEE Xplore. Restrictions apply.

[10] X. Cao, H. Lin, S. Guo, T. Xiong, and L. Jiao, "Transformer-based masked autoencoder with contrastive loss for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524312.

[11] S. K. Roy, A. Sukul, A. Jamali, J. M. Haut, and P. Ghamisi, "Cross hyperspectral and LiDAR attention transformer: An extended self-attention for land use and land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512815, doi: 10.1109/TGRS.2024.3374324.

[12] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022, doi: 10.1109/TIP.2022.3162964.

[13] H. Wu, S. Dai, C. Liu, A. Wang, and Y. Iwahori, "A novel dual-encoder model for hyperspectral and LiDAR joint classification via contrastive learning," *Remote Sens.*, vol. 15, no. 4, p. 924, Feb. 2023.

[14] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[15] Y. Xu, D. Wang, L. Zhang, and L. Zhang, "Dual selective fusion transformer network for hyperspectral image classification," *Neural Netw.*, vol. 187, Jul. 2025, Art. no. 107311.

[16] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005, doi: 10.1109/TGRS.2004.842478.

[17] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Land-cover classification using both hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 6, no. 3, pp. 189–215, Jul. 2015, doi: 10.1080/19479832.2015.1055833.

[18] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017, doi: 10.1109/JSTARS.2016.2634863.

[19] J. Xia, N. Yokoya, and A. Iwasaki, "A novel ensemble classifier of hyperspectral and LiDAR data using morphological features," *presented at the IEEE Int. Conf. Acoust., New Orleans, LA, USA*, Mar. 2017, doi: 10.1109/ICASSP.2017.7953345.

[20] X. Xu, J. Li, and A. Plaza, "Fusion of hyperspectral and LiDAR data using morphological component analysis," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3575–3578, doi: 10.1109/IGARSS.2016.7729926.

[21] J. Xia, W. Liao, and P. Du, "Hyperspectral and LiDAR classification with semisupervised graph fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 4, pp. 666–670, Apr. 2020, doi: 10.1109/LGRS.2019.2928009.

[22] H. Jing et al., "Aggregative and contrastive dual-view graph attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530817.

[23] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[24] P. Zhong, Z. Gong, S. Li, and C.-B. Schönlieb, "Learning to diversify deep belief networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3516–3530, Jun. 2017.

[25] W. Yu, L. Gao, H. Huang, Y. Shen, and G. Shen, "HI^{2}D^{2}FNet: Hyperspectral intrinsic image decomposition guided data fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521715.

[26] L. Shutao, L. Congyu, and K. Xudong, "Development status and future prospects of multi-source remote sensing image fusion," *Nat. Remote Sens. Bull.*, vol. 25, no. 1, pp. 148–166, 2021.

[27] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3017414.

[28] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[29] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-resnet and the impact of residual connections on learning," in *Proc. AAAI Conf. Artif. Intell.*, 2017, vol. 31, no. 1, pp. 1–7.

[30] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[31] B. Fang, Y. Liu, H. Zhang, and J. He, "Hyperspectral image classification based on 3D asymmetric inception network with data fusion transfer learning," *Remote Sens.*, vol. 14, no. 7, p. 1711, Apr. 2022, doi: 10.3390/rs14071711.

[32] X. Cai, Q. Lai, Y. Wang, W. Wang, Z. Sun, and Y. Yao, "Poly kernel inception network for remote sensing detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27706–27716.

[33] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[34] Y. Deng, Y. Cao, S. Chen, and X. Cheng, "Residual attention network with atrous spatial pyramid pooling for soil element estimation in LUCAS hyperspectral data," *Appl. Sci.*, vol. 15, no. 13, p. 7457, Jul. 2025.

[35] P. Wu et al., "Research on water extraction from high resolution remote sensing images based on deep learning," *Frontiers Remote Sens.*, vol. 4, Dec. 2023, Art. no. 1283615.

[36] R. Fang, "Application of deep learning to land cover classification: Practical issues and strategies," M.S. thesis, Dept. Urban Stud. Planning, Massachusetts Inst. Technol., Cambridge, MA, USA, 2023.

[37] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[38] X. Li, M. Ding, and A. Pižurica, "Deep feature fusion via two-stream convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2615–2629, Apr. 2020, doi: 10.1109/TGRS.2019.2952758.

[39] Y. Jiang, Y. Gu, and X. Li, "Pillar-voxel fusion network for 3D object detection in airborne hyperspectral point clouds," *Sci. China Inf. Sci.*, vol. 69, no. 1, Jan. 2026, Art. no. 112301, doi: 10.1007/s11432-024-4458-0.

[40] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020, doi: 10.1109/TGRS.2020.2982064.

[41] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021, doi: 10.1109/TGRS.2020.3016820.

[42] E. K. Ghasrodashti, P. Adibi, H. Karshenas, H. B. Kashani, and J. Chanussot, "Multimodal image classification based on convolutional network and attention-based hidden Markov random field," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511114, doi: 10.1109/TGRS.2025.3560913.

[43] K. M. Hussain, K. Zhao, Y. Zhou, A. Ali, and Y. Li, "Cross attention based dual-modality collaboration for hyperspectral image and LiDAR data classification," *Remote Sens.*, vol. 17, no. 16, p. 2836, Aug. 2025. [Online]. Available: https://www.mdpi.com/2072-4292/17/16/2836

[44] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716, doi: 10.1109/TGRS.2022.3232498.

[45] W. Wang et al., "Dual-branch feature fusion network based cross-modal enhanced CNN and transformer for hyperspectral and LiDAR classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[46] Z. Li, J. Wu, Y. Zhang, and Y. Yan, "CMFNet: Cross mamba fusion network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4418614, doi: 10.1109/TGRS.2025.3610348.

[47] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[48] G. Papandreou, I. Kokkinos, and P.-A. Savalle, "Modeling local and global deformations in deep learning: Epitomic convolution, multiple instance learning, and sliding window detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 390–399.

[49] G. Lin, C. Shen, A. van den Hengel, and I. Reid, "Efficient piecewise training of deep structured models for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3194–3203.

[50] M. Liu, L. Jiao, X. Liu, L. Li, F. Liu, and S. Yang, "C-CNN: Contourlet convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2636–2649, Jun. 2021.

[51] J. Gao, L. Jiao, F. Liu, S. Yang, B. Hou, and X. Liu, "Multiscale curvelet scattering network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3665–3679, Jul. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:03:12 UTC from IEEE Xplore. Restrictions apply.

[52] J. Gao et al., "Multiscale dynamic curvelet scattering network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 7999–8012, Jun. 2024.
[53] C. Farabet, C. Couprie, L. Najman, and Y. LeCun, "Learning hierarchical features for scene labeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1915–1929, Aug. 2013.
[54] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6053–6062.
[55] H. Li, H. Pan, X. Liu, J. Ren, Z. Du, and J. Cao, "GLVMamba: A global–local visual state-space model for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4412115, doi: 10.1109/TGRS.2025.3572127.
[56] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.
[57] W. Liu, A. Rabinovich, and A. C. Berg, "ParseNet: Looking wider to see better," 2015, *arXiv:1506.04579*.
[58] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[59] G. Zhang, Z. Li, C. Tang, J. Li, and X. Hu, "CEDNet: A cascade encoder–decoder network for dense prediction," *Pattern Recognit.*, vol. 158, Feb. 2025, Art. no. 111072.
[60] Y. Cao, J. Bin, J. Hamari, E. Blasch, and Z. Liu, "Multimodal object detection by channel switching and spatial attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 403–411.
[61] L. Breiman, *Random Forests*. Cham, Switzerland: Springer, 2001, pp. 5–32.



Minghui Wu is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

Her research interests include light detection and ranging (LiDAR) remote sensing, quantitative remote sensing, spatial heterogeneity modeling, and forest carbon sink monitoring and mapping.



Shuo Shi is currently a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China. He is committed to the research of new light detection and ranging (LiDAR) technology and quantitative remote sensing application of vegetation, mainly engaged in hyperspectral, laser-induced fluorescence, and 3-D information integration acquisition, LiDAR system design and development, quantitative inversion of vegetation parameters and carbon sink estimation, target recognition and classification based on deep learning, and 3-D color reconstruction.



Youchai Lin received the B.E. degree in electronic information science and technology from the School of Electronics Science and Technology, Xiamen University, Xiamen, China, in 2023. He is currently pursuing the M.S. degree with the School of Electronic Information, Wuhan University, Wuhan, China.

His major research interests include deep learning methods and hyperspectral light detection and ranging (LiDAR).



Deping Chu is currently pursuing the Ph.D. degree with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China.

His research interests include geoscience big data mining, geoscience knowledge graph, and GIS modeling.



Jiayun Niu received the Bachelor of Engineering degree in surveying and mapping engineering from Shandong Jianzhu University, Jinan, China, in 2020 and the master's degree from China University of Geosciences, Wuhan, China, in 2024. She is currently pursuing the Ph.D. degree with the School of Electronic Information, Wuhan University, Wuhan.

Her research interests include light detection and ranging (LiDAR), vegetation carbon storage monitoring, and new energy.



Xingtao Tang received the B.S. degree in measurement and control technology and instrumentation from the School of Mechanical and Electronic Engineering, Wuhan University of Technology, Wuhan, China, in 2020, and the master's degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, in 2024. He is currently pursuing the Ph.D. degree with the School of Electronic Information, Wuhan University.

His research interests include hyperspectral light detection and ranging (LiDAR) data processing, point cloud classification, and vegetation carbon storage monitoring.



Wei Gong is currently a Professor and a Doctoral Supervisor with Wuhan University, Wuhan, China. He has long been committed to the research of laser remote sensing technology and methods, including the key core technology of photoelectric detection, laser remote sensing system integration method, and satellite remote sensing data science application research. His scientific research focuses on the technology and application of optical and laser remote sensing, mainly focusing on new air-to-ground laser remote sensing technology, optical and laser remote sensing data processing, and optical and laser spatial information quantitative application.

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